

Theme 2: Limits, Continuity & Derivatives.

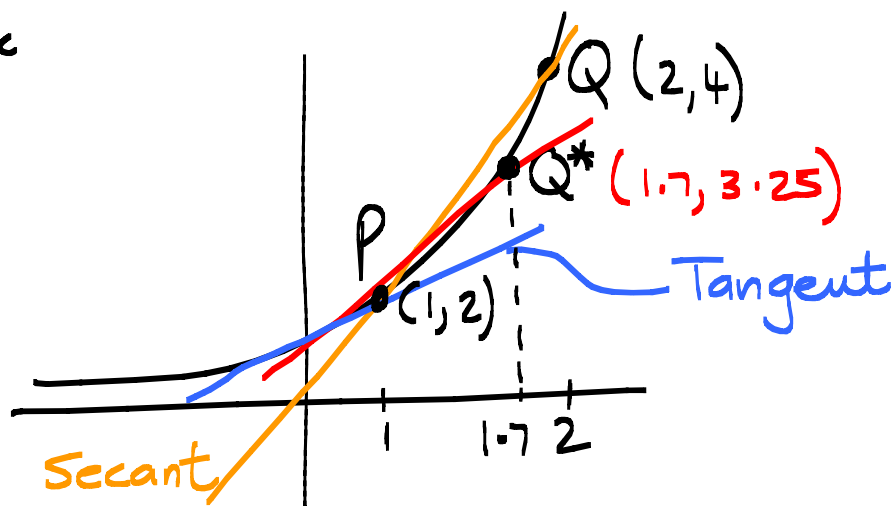
Note Title

2014/02/19

Unit 2.1 Tangent & Velocity problems. p 78-82

Introduction - how limits arise when we try to find the tangent to a curve or velocity of an object.

Sketch $y = 2^x$



To find the equation of the secant PQ through points (1, 2) and (2, 4).

$$m_{PQ} = \frac{4-2}{2-1} = 2$$

and $\rightarrow y - y_1 = m(x - x_1)$
 $y - 2 = \underline{2}(x - 1)$
 $y = 2x$

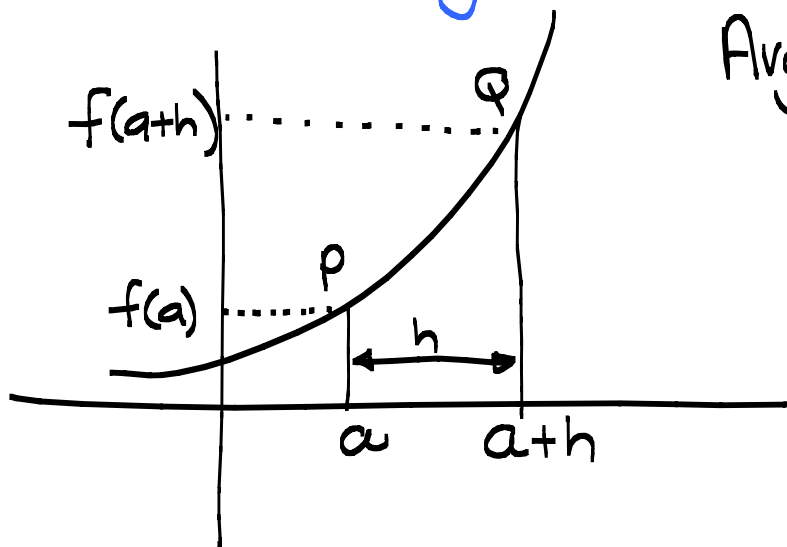
Move Q to $Q^* (1.7, 3.25)$ now equation PQ* becomes

$$y = \underline{1.79}x + 0.21$$
$$m_{PQ^*} = 1.79$$

Moving Q to Q^* etc. closer and closer to P will result in the line becoming the tangent line to the curve in the point P .

We need to find the slope of this tangent line to find the equation. It so happens that it is 1.387 but we are going to need calculus to find it.

Average rate of change: $\frac{x_2 - x_1}{y_2 - y_1}$
 Slope of tangent is instantaneous rate of change.



$$\begin{aligned} \text{Avg} &= \frac{f(a+h) - f(a)}{a+h - a} \\ &= \frac{f(a+h) - f(a)}{h} \end{aligned}$$

By making h smaller and smaller we get

Instant. rate of change

$$= \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

↙
 This is why we need to know how to handle the LIMIT of a function!